

Compactness in Metric Spaces

MA 511 — Real Analysis

Definition

A subset K of a metric space X is compact if every open cover of K has a finite subcover.

Sequential compactness

In a metric space, compactness is equivalent to sequential compactness: every sequence in K has a subsequence converging to a point in K .

Heine–Borel

For $K \subset \mathbb{R}^n$: K is compact if and only if K is closed and bounded. This equivalence does NOT hold in general metric spaces.